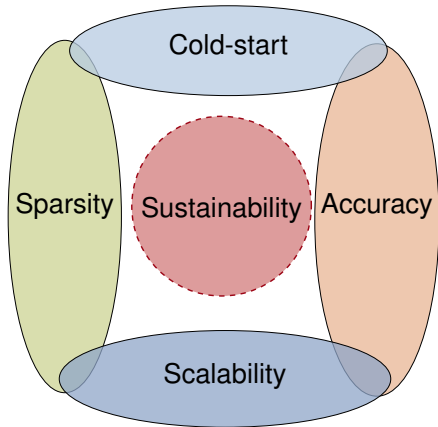


Energy-Aware Hybrid Recommender Systems Across the User Lifecycle

Saina Amiri Moghadam

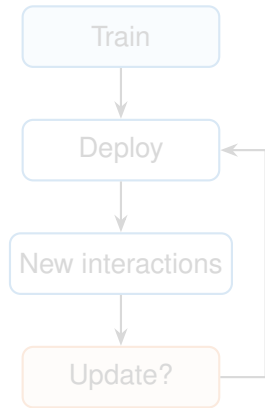
Chair of Connected Mobility
TUM School of Computation, Information and
Technology
Technical University of Munich

October 6, 2026



Introduction

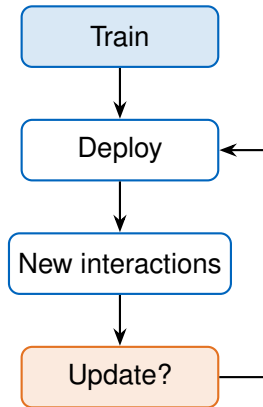
- Recommender models are becoming more expensive without proportionate gains in quality¹
- A recommender is not trained once: new users arrive and preferences shift, so the model must be kept up to date
- The recurring carbon cost of keeping it current has not been measured



¹G. Spillo et al., in Proceedings of the 17th acm conference on recommender systems, RecSys '23 (2023), pp. 856–862
T. Vente et al., in Proceedings of the 18th acm conference on recommender systems, RecSys '24 (2024), pp. 580–590

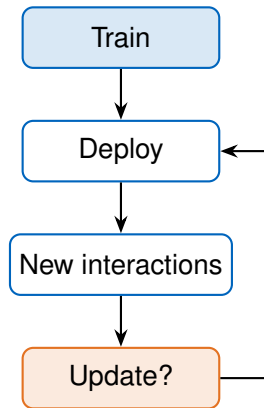
Introduction

- Recommender models are becoming more expensive without proportionate gains in quality¹
- A recommender is not trained once: new users arrive and preferences shift, so the model must be kept up to date
- The recurring carbon cost of keeping it current has not been measured



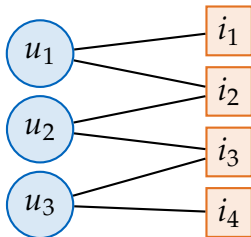
¹ G. Spillo et al., in Proceedings of the 17th acm conference on recommender systems, RecSys '23 (2023), pp. 856–862
T. Vente et al., in Proceedings of the 18th acm conference on recommender systems, RecSys '24 (2024), pp. 580–590

- Recommender models are becoming more expensive without proportionate gains in quality¹
- A recommender is not trained once: new users arrive and preferences shift, so the model must be kept up to date
- The recurring carbon cost of keeping it current has not been measured



¹G. Spillo et al., in Proceedings of the 17th acm conference on recommender systems, RecSys '23 (2023), pp. 856–862
T. Vente et al., in Proceedings of the 18th acm conference on recommender systems, RecSys '24 (2024), pp. 580–590

What does it cost to keep a recommender up to date, and can a cheaper hybrid mechanism deliver comparable quality?



User–Item Interaction Graph

Neighborhood aggregation:

$$\mathbf{e}_u^{(k)} = \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u|} \sqrt{|\mathcal{N}_i|}} \mathbf{e}_i^{(k-1)}$$

$$\text{Prediction: } \hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$$

¹X. He et al., in Proceedings of the 43rd international acm sigir conference on research and development in information retrieval, SIGIR '20 (2020), pp. 639–648

CPU and RAM energy estimated
with CodeCarbon

$$\text{emissions} = C \cdot E$$

$$C = \sum_{s \in S} e_s \cdot p_s$$

$$C \text{ (Germany)} = 381 \text{ g CO}_2\text{eq/kWh}$$

¹CodeCarbon, *Methodology*, (2024) <https://docs.codecarbon.io/latest/explanation/methodology/>

CPU and RAM energy estimated
with CodeCarbon

$$\text{emissions} = C \cdot E$$

$$C = \sum_{s \in S} e_s \cdot p_s$$

$$C \text{ (Germany)} = 381 \text{ g CO}_2\text{eq/kWh}$$

¹CodeCarbon, *Methodology*, (2024) <https://docs.codecarbon.io/latest/explanation/methodology/>

CPU and RAM energy estimated
with CodeCarbon

$$\text{emissions} = C \cdot E$$

$$C = \sum_{s \in S} e_s \cdot p_s$$

$$C \text{ (Germany)} = 381 \text{ g CO}_2\text{eq/kWh}$$

¹CodeCarbon, *Methodology*, (2024) <https://docs.codecarbon.io/latest/explanation/methodology/>

Evaluation Metrics

Per batch: score all items,
mask items already seen,
recommend the top 10

Relevant: items the user
liked in this batch

$$\text{Precision@10} = \frac{\text{relevant items retrieved}}{10}$$

$$\text{Recall@10} = \frac{\text{relevant items retrieved}}{\text{relevant items}}$$

$$\text{NDCG@10} = \frac{\text{hits, weighted by rank}}{\text{best possible ranking}}$$

¹ C. D. Manning et al., *Introduction to information retrieval*, (Cambridge University Press, 2008)

Evaluation Metrics

Per batch: score all items,
mask items already seen,
recommend the top 10

Relevant: items the user
liked in this batch

$$\text{Precision@10} = \frac{\text{relevant items retrieved}}{10}$$

$$\text{Recall@10} = \frac{\text{relevant items retrieved}}{\text{relevant items}}$$

$$\text{NDCG@10} = \frac{\text{hits, weighted by rank}}{\text{best possible ranking}}$$

¹ C. D. Manning et al., *Introduction to information retrieval*, (Cambridge University Press, 2008)

Evaluation Metrics

Per batch: score all items,
mask items already seen,
recommend the top 10

Relevant: items the user
liked in this batch

$$\text{Precision@10} = \frac{\text{relevant items retrieved}}{10}$$

$$\text{Recall@10} = \frac{\text{relevant items retrieved}}{\text{relevant items}}$$

$$\text{NDCG@10} = \frac{\text{hits, weighted by rank}}{\text{best possible ranking}}$$

¹ C. D. Manning et al., *Introduction to information retrieval*, (Cambridge University Press, 2008)

Evaluation Metrics

Per batch: score all items,
mask items already seen,
recommend the top 10

Relevant: items the user
liked in this batch

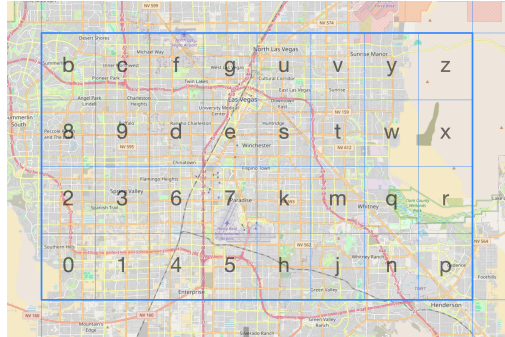
$$\text{Precision@10} = \frac{\text{relevant items retrieved}}{10}$$

$$\text{Recall@10} = \frac{\text{relevant items retrieved}}{\text{relevant items}}$$

$$\text{NDCG@10} = \frac{\text{hits, weighted by rank}}{\text{best possible ranking}}$$

¹C. D. Manning et al., *Introduction to information retrieval*, (Cambridge University Press, 2008)

Content Signal: Geohash



G. Niemeyer, *Geohash.org is public!*, (2008)

<https://web.archive.org/web/20080305223755/http://blog.labix.org/#post-84>

Incremental Update:

Keeps trained users up to
date

Content-Based Initialization:

Places users the model has
never seen



Combine both and compare quality against carbon
emissions

Incremental Update:

Keeps trained users up to
date

Content-Based Initialization:

Places users the model has
never seen



Combine both and compare quality against carbon
emissions

Methodology

- Split Yelp and MovieLens along the global timeline: 80% training, 20% streamed in batches
- Compare no-update, incremental, content-based initialization, their combination and full retraining
- Measure Recall@10 per batch and emissions per phase with CodeCarbon

Methodology

- Split Yelp and MovieLens along the global timeline: 80% training, 20% streamed in batches
- Compare no-update, incremental, content-based initialization, their combination and full retraining
- Measure Recall@10 per batch and emissions per phase with CodeCarbon

Methodology

- Split Yelp and MovieLens along the global timeline: 80% training, 20% streamed in batches
- Compare no-update, incremental, content-based initialization, their combination and full retraining
- Measure Recall@10 per batch and emissions per phase with CodeCarbon

1 Profile trained users

	g_1	g_2	g_3	g_4
u_1	3	0	1	0
u_2	0	2	0	1
u_3	1	0	4	0
u_4	0	0	1	2

check-ins per geohash
(same for categories)

Content-Based Initialization

1 Profile trained users

	g_1	g_2	g_3	g_4
u_1	3	0	1	0
u_2	0	2	0	1
u_3	1	0	4	0
u_4	0	0	1	2

check-ins per geohash
(same for categories)

2 Score similarity

new user	1	0	2	0
u_1				0.67
u_2				0.00
u_3				1.00
u_4				0.33

min-sum overlap
 $S = 0.7 \cdot \text{geo} + 0.3 \cdot \text{cat}$

Content-Based Initialization

1 Profile trained users

	g_1	g_2	g_3	g_4
u_1	3	0	1	0
u_2	0	2	0	1
u_3	1	0	4	0
u_4	0	0	1	2

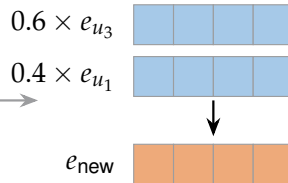
check-ins per geohash
(same for categories)

2 Score similarity

new user	1	0	2	0
u_1	0.67			
u_2	0.00			
u_3	1.00			
u_4	0.33			

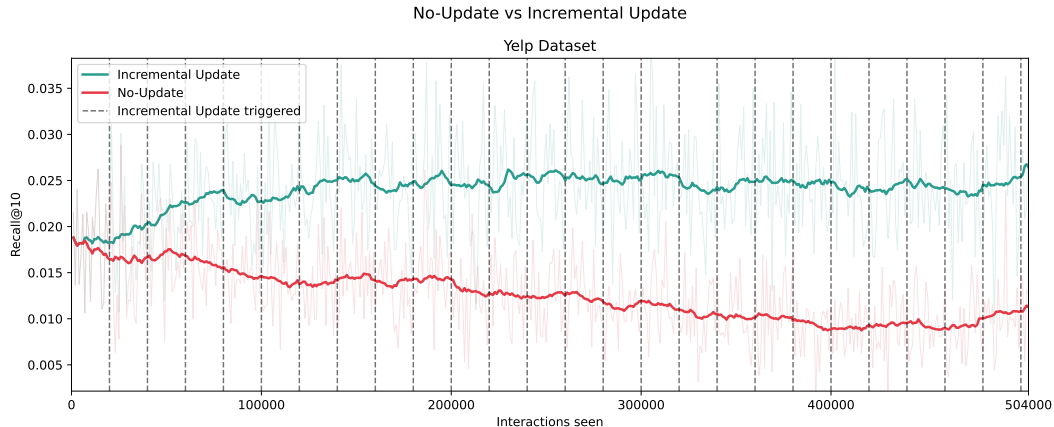
min-sum overlap
 $S = 0.7 \cdot \text{geo} + 0.3 \cdot \text{cat}$

3 Initialize embedding

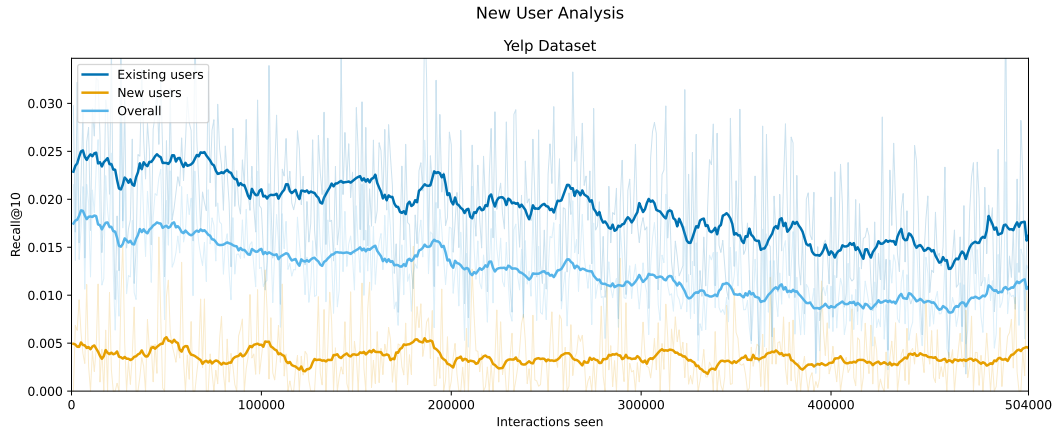


weights = normalized scores
of the top- k users ($k = 20$)

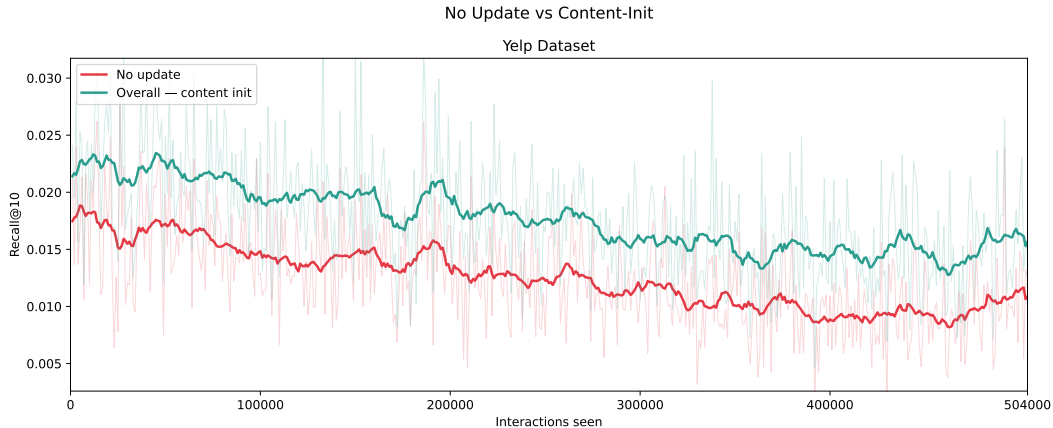
Yelp: No-Update vs. Incremental Update



Yelp: Existing vs. New Users

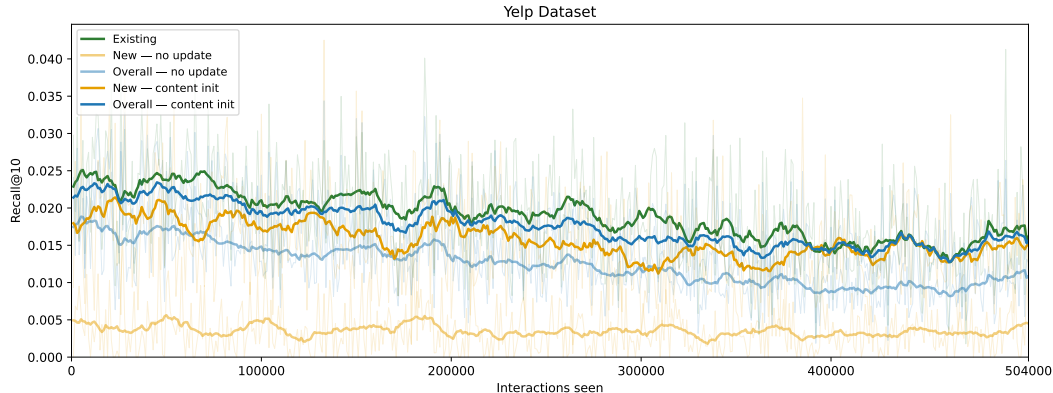


Yelp: Content-Based Initialization



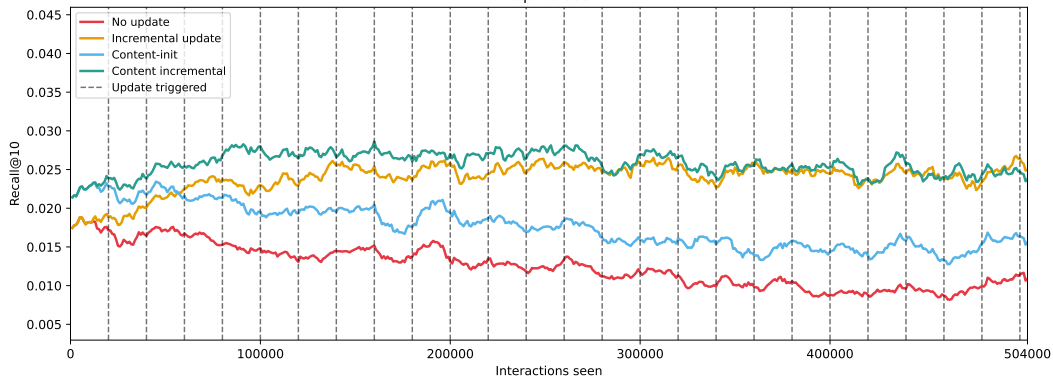
Yelp: Content-Based Initialization

No Update vs Content-Init by User Group

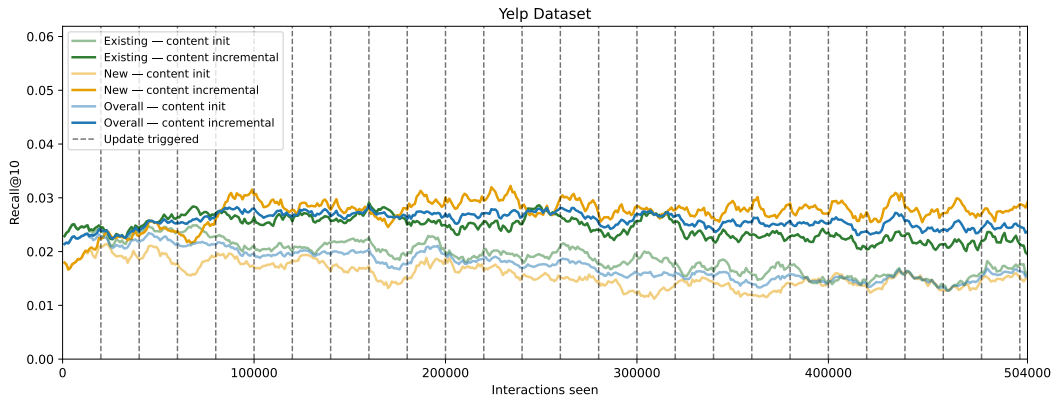


Strategy Comparison

Yelp Dataset



Content-Init vs Content Incremental by User Group



	No-Update	Incremental	Content-Init	Content + Incremental
Recall@10	0.0125	+91%	+41%	+106%
Emissions (% of training)	5.20%	6.82%	6.14%	7.55%

Mean Recall@10 (relative to no-update) and streaming emissions as a share of the one-time training cost on Yelp

	No-Update	Incremental	Content-Init	Content + Incremental
Recall@10	0.0125	+91%	+41%	+106%
Emissions (% of training)	5.20%	6.82%	6.14%	7.55%

Mean Recall@10 (relative to no-update) and streaming emissions as a share of the one-time training cost on Yelp

Comparison of Strategies

	No-Update	Incremental	Content-Init	Content + Incremental
Recall@10	0.0125	+91%	+41%	+106%
Emissions (% of training)	5.20%	6.82%	6.14%	7.55%

Mean Recall@10 (relative to no-update) and streaming emissions as a share of the one-time training cost on Yelp

	No-Update	Incremental	Content-Init	Content + Incremental
Recall@10	0.0125	+91%	+41%	+106%
Emissions (% of training)	5.20%	6.82%	6.14%	7.55%

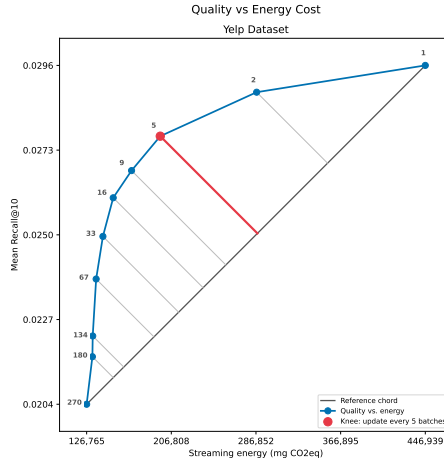
Mean Recall@10 (relative to no-update) and streaming emissions as a share of the one-time training cost on Yelp

Comparison of Strategies

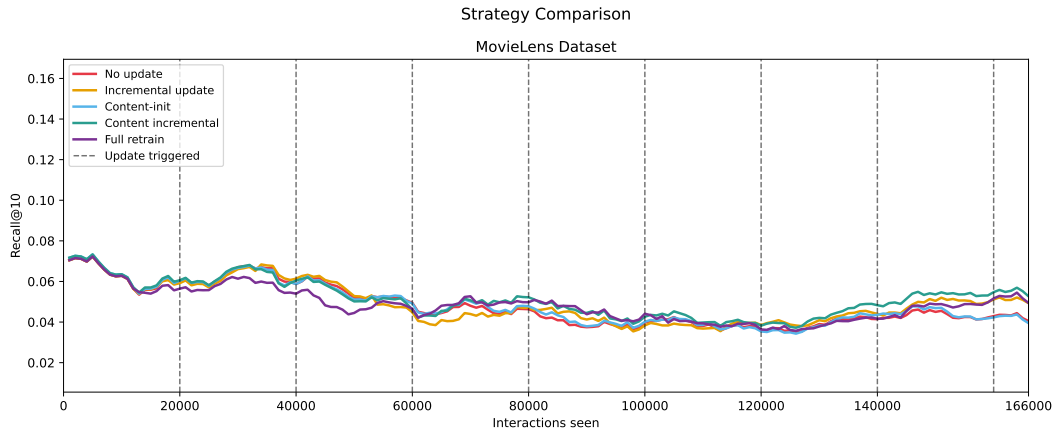
	No-Update	Incremental	Content-Init	Content + Incremental
Recall@10	0.0125	+91%	+41%	+106%
Emissions (% of training)	5.20%	6.82%	6.14%	7.55%

Mean Recall@10 (relative to no-update) and streaming emissions as a share of the one-time training cost on Yelp

Yelp: Update Frequency vs. Emissions

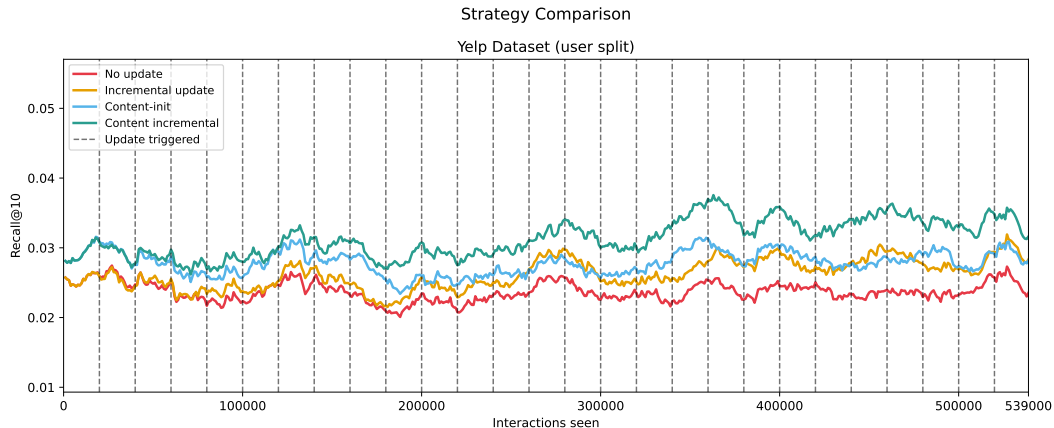


¹V. Satopää et al., in 2011 31st international conference on distributed computing systems workshops (2011), pp. 166–171



¹F. M. Harper and J. A. Konstan, ACM Transactions on Interactive Intelligent Systems 5, 19 (2015)

Pitfall: Per-User Split



¹Y. Ji et al., ACM Transactions on Information Systems **41**, 75, 1 (2023)

Pitfall: Per-User Split



¹Y. Ji et al., ACM Transactions on Information Systems **41**, 75, 1 (2023)

Cheap updates recover most of the lost quality at a fraction of the training cost, but only where the data has something to correct



Diagnose the source of decay before choosing an update strategy

Cheap updates recover most of the lost quality at a fraction of the training cost, but only where the data has something to correct



Diagnose the source of decay before choosing an update strategy

Next Steps

- Causal incremental graph convolution (CI-LightGCN), measured in energy rather than time²
- Windowed graph construction to bound the update cost³
- Streams with controlled, documented drift⁴
- More datasets with less strict interaction filtering

²S. Ding et al., IEEE Transactions on Neural Networks and Learning Systems **35**, 4718 (2024)

³A. Ghiye et al., in Proceedings of the 5th acm international conference on ai in finance, ICAIF '24 (2024), pp. 231–238

⁴V. M. A. Souza et al., Data Mining and Knowledge Discovery **34**, 1805 (2020)

Next Steps

- Causal incremental graph convolution (CI-LightGCN), measured in energy rather than time²
- Windowed graph construction to bound the update cost³
- Streams with controlled, documented drift⁴
- More datasets with less strict interaction filtering

²S. Ding et al., IEEE Transactions on Neural Networks and Learning Systems **35**, 4718 (2024)

³A. Ghiye et al., in Proceedings of the 5th acm international conference on ai in finance, ICAIF '24 (2024), pp. 231–238

⁴V. M. A. Souza et al., Data Mining and Knowledge Discovery **34**, 1805 (2020)

Next Steps

- Causal incremental graph convolution (CI-LightGCN), measured in energy rather than time²
- Windowed graph construction to bound the update cost³
- Streams with controlled, documented drift⁴
- More datasets with less strict interaction filtering

²S. Ding et al., IEEE Transactions on Neural Networks and Learning Systems **35**, 4718 (2024)

³A. Ghiye et al., in Proceedings of the 5th acm international conference on ai in finance, ICAIF '24 (2024), pp. 231–238

⁴V. M. A. Souza et al., Data Mining and Knowledge Discovery **34**, 1805 (2020)

Next Steps

- Causal incremental graph convolution (CI-LightGCN), measured in energy rather than time²
- Windowed graph construction to bound the update cost³
- Streams with controlled, documented drift⁴
- More datasets with less strict interaction filtering

²S. Ding et al., IEEE Transactions on Neural Networks and Learning Systems **35**, 4718 (2024)

³A. Ghiye et al., in Proceedings of the 5th acm international conference on ai in finance, ICAIF '24 (2024), pp. 231–238

⁴V. M. A. Souza et al., Data Mining and Knowledge Discovery **34**, 1805 (2020)

Thank you!

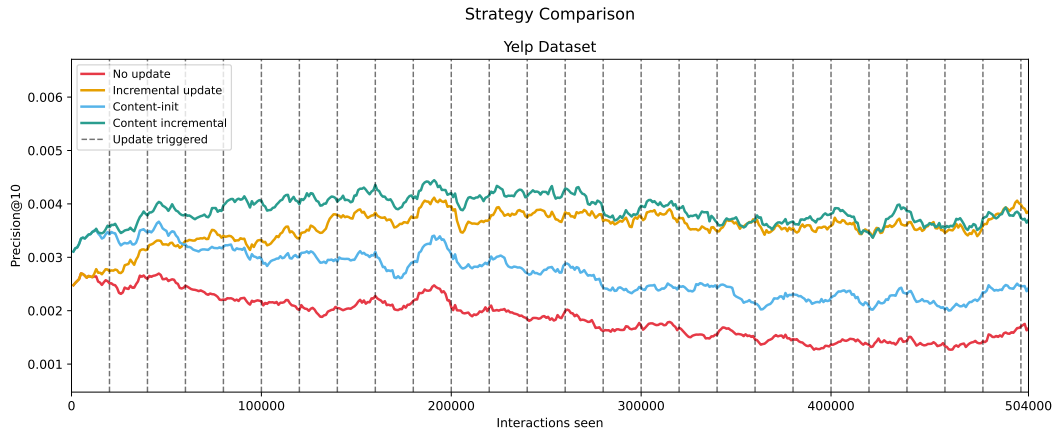
Chair of Connected Mobility

Examiner: Prof. Dr.-Ing. Jörg Ott

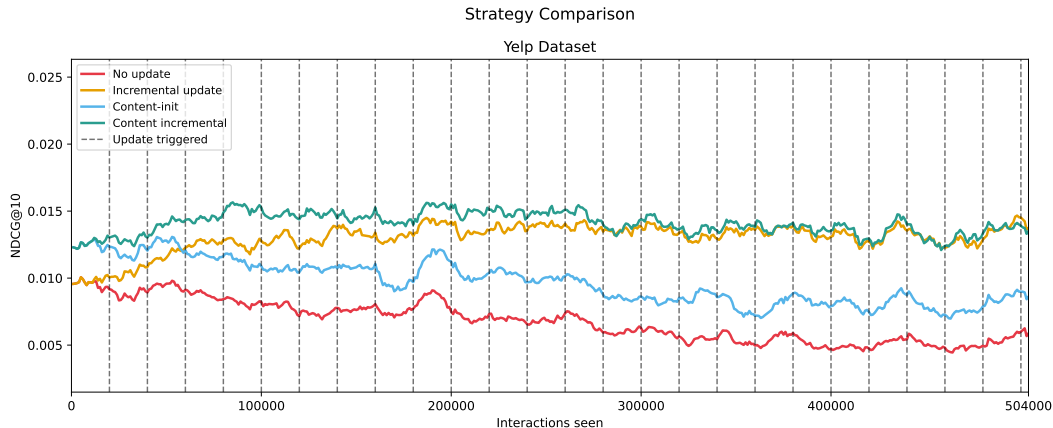
Advisor: Prof. Wolfgang Wörndl

Appendix

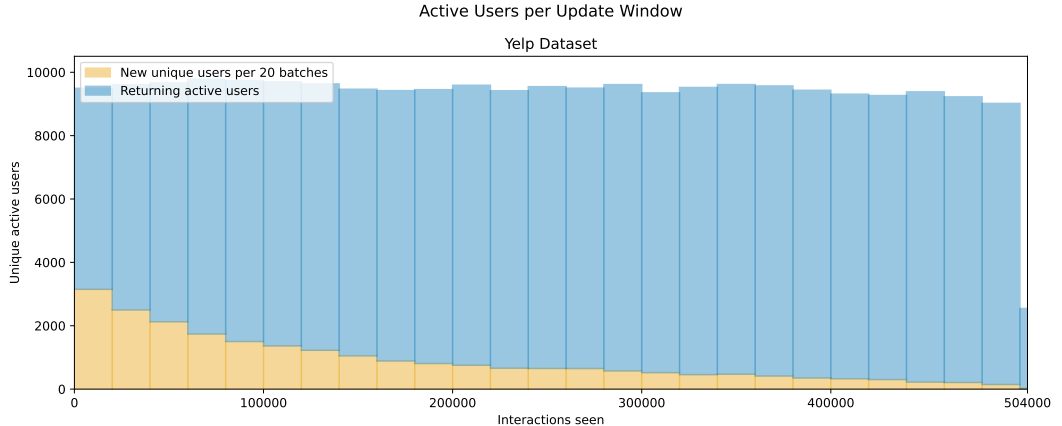
Yelp: Precision@10, All Strategies



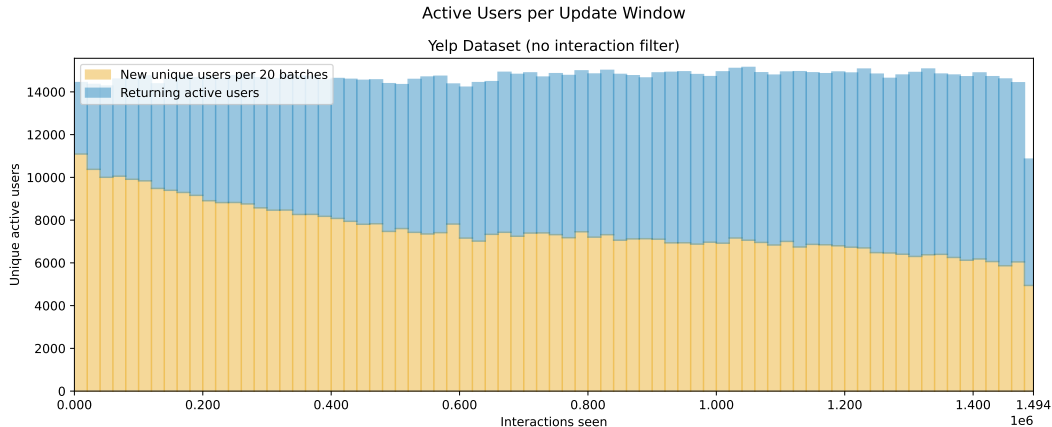
Yelp: NDCG@10, All Strategies



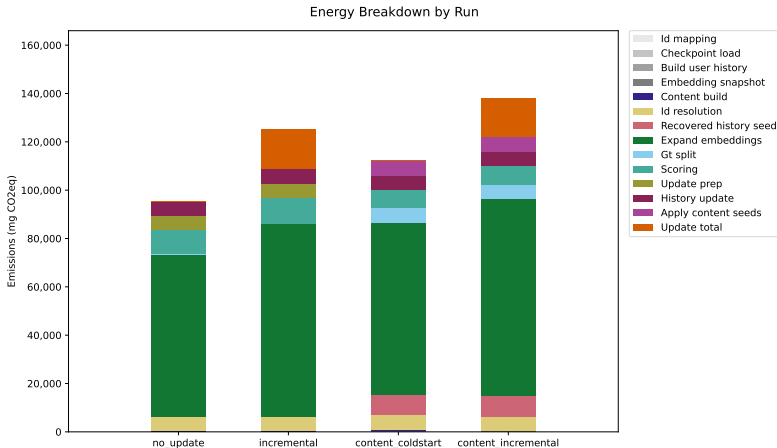
Yelp: New Unique Users per Update Window



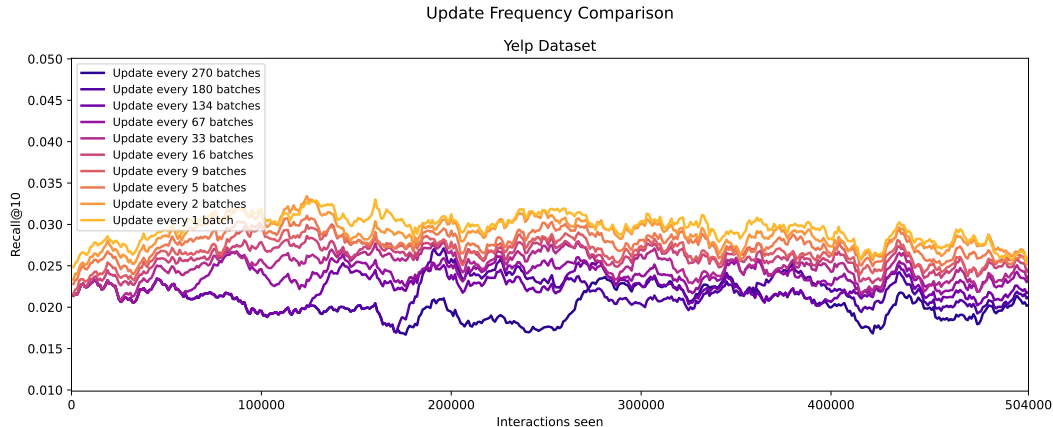
Yelp: New Unique Users per Update Window



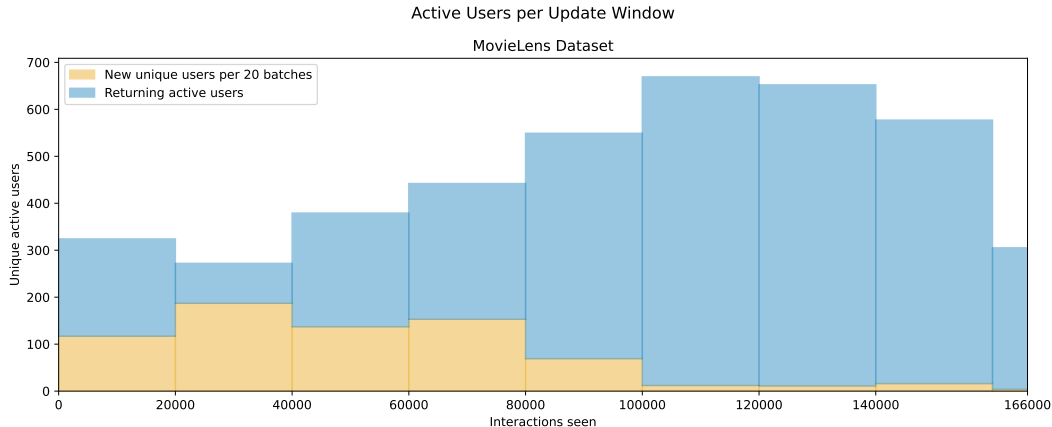
Yelp: Emissions by Phase



Yelp: Update Frequency Variants



MovieLens: New Unique Users per Update Window



Carbon Cost per Strategy

	No-Update	Incremental	Content-Init	Content + Incremental
Yelp streaming	95,306	125,053	112,422	138,321
Yelp training		1,832,436		
MovieLens streaming	17,286	18,654	20,045	22,618
MovieLens training		188,839		

Carbon emissions in mg CO₂eq. Full retrain on MovieLens: 1,387,568 streaming, 792× incremental per update.